Labwork 1 - Gradient Descent

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1 Gradient Descent Implementation

This labwork is for implementing gradient descent algorithm from scratch.
In this labwork, I will work on $f(x) = x^2$ and the goal is to find the local minimum of the function. For the setup, I will experiment the algorithm with the initial value is $x_0 = 5$. The core update rule of the algorithm is defined as below:

$$x_{n+1} = x_n - lr \times f'(x_n)$$

where:

- x_n: is the current value.
- x_{n+1}: is the update value.
- lr: is the learning rate.
- $f'(x)$: is the first order derivative at x_n.

As the choice of learning rate has large effect on the performance of the algorithm, I have tried 4 different learning rates, which are 0.001, 0.005, 0.1, 0.5. Below is the convergence speed of each learning rate:

Learning rate	**Converged iteration**
0.001	186901
0.005	37231
0.1	1677
0.5	1

The table 1 shows that the convergence speed is highly dependent on the learning rate, with smaller values like 0.001 and 0.005 requiring an large number of iterations. In contrast, $lr = 0.5$ is the most efficient in this case, it achieve the local minimum in a single step.